

Lecture 1: Data & Image Representation (MCQs)

1. _____ is called a 1-bit monochrome image since it contains no color.

- a) Gray image b) Binary image c) RGB image d) CMY image

• Answer: b) Binary image,

2. A group of 4 bits is called a _____.

- a) Byte b) Word c) Nibble d) Double Word

• Answer: c) Nibble

3. In a Gray image, a pixel is represented by _____ bits.

- a) 1 b) 4 c) 8 d) 24

• Answer: c) 8

4. The range of gray values in an 8-bit image is between _____.

- a) 0 and 127 b) 0 and 255 c) 0 and 65535 d) -128 and 127

• Answer: b) 0 and 255

5. A 640 x 480 grayscale image requires _____ of storage.

- a) 300 KB b) 921.6 KB c) 150 KB d) 1.2 MB

• Answer: a) 300 KB,

6. Which type of image supports 16,777,216 possible combined colors?

- a) Binary Image b) Gray Image c) Color Image (24-bit) d) Monochrome Image

• Answer: c) Color Image (24-bit)

7. Without compression, a 640x480 24-bit color image would require _____ of storage.

- a) 300 KB b) 600 KB c) 921.6 KB d) 1024 KB

• Answer: c) 921.6 KB

8. Sound is an _____ acoustic wave.

- a) Digital b) Analog c) Binary d) Discrete

• Answer: b) Analog

9. In sound representation, the larger the _____, the louder the sound.

- a) Frequency b) Amplitude c) Phase d) Sampling rate

• Answer: b) Amplitude

10. By reducing the sound wave sample rate, the amount of data you need to store is _____. a) reduced b) increased c) compressed d) decoded

• Answer: a) reduced (Question directly from Exam 2025)

11. The typical sampling rate for digital sound is _____ per second.

- a) 20 kHz b) 44 kHz c) 48 kHz d) 50 kHz

• **Answer:** c) 48 kHz

12. In video representation, each image in the sequence is called a _____.

- a) Shot b) Pixel c) Frame d) Sample

• **Answer:** c) Frame

13. To create the impression of a continuous movie, frames are flashed on the screen faster than _____. a) 24/sec b) 30/sec c) 50/sec d) 60/sec

• **Answer:** c) 50/sec

14. ASCII stands for _____.

- a) American Standard Code for Information Interchange
b) American System Code for Information Interchange
c) Arab Standard Code for Information Interchange
d) American Standard Code for Internet Interchange

• **Answer:** a) American Standard Code for Information Interchange

15. Standard ASCII is limited to _____ possible combinations.

- a) 128 b) 256 c) 512 d) 65536

• **Answer:** b) 256

16. Which code was used in the past by IBM mainframes?

- a) ASCII b) Unicode c) EBCDIC d) UTF-8

• **Answer:** c) EBCDIC

17. The decimal ASCII value for the character 'A' is _____.

- a) 65 b) 97 c) 48 d) 32

• **Answer:** a) 65,

18. The binary representation of the character 'A' (65) is _____.

- a) 01000001 b) 01100001 c) 00100000 d) 00110001

• **Answer:** a) 01000001

19. The decimal ASCII value for the character 'a' (small) is _____.

- a) 65 b) 96 c) 97 d) 100

• **Answer:** c) 97

20. The decimal ASCII value for 'Space' is _____.

- a) 0 b) 10 c) 32 d) 255

• **Answer:** c) 32

21. Unicode uses _____ bytes for each character.

a) 1 b) 2 c) 3 d) 4

• **Answer:** b) 2

22. A major disadvantage of Unicode compared to ASCII is that it _____.

a) Supports fewer characters b) Is not standard c) Takes twice as much space d) Cannot represent English

• **Answer:** c) Takes twice as much space

23. Unicode allows _____ different symbols to be represented.

a) 256 b) 4096 c) 65,536 d) 1,048,576

• **Answer:** c) 65,536

24. In UTF-8, if a character takes 3 bytes, the first byte starts with _____.

a) 10xxxxxx b) 110xxxxx c) 1110xxxx d) 11110xxx

• **Answer:** c) 1110xxxx

25. In a multi-byte UTF-8 character, every byte after the first one starts with _____. a) 00 b) 01 c) 10 d) 11

• **Answer:** c) 10

26. Code points from U+0080 to U+07FF in UTF-8 are represented using _____ bytes. a) 1 b) 2 c) 3 d) 4

• **Answer:** b) 2

27. Is 10000000 a valid start byte for a 2-byte UTF-8 character? a) Yes b) No, it should start with 110 c) No, it should start with 0 d) No, it represents ASCII

• **Answer:** b) No, it should start with 110

28. The hexadecimal equivalent of (233)₁₀ is _____. (Similar to Exam Q4)

a) E9 b) A5 c) 9E d) E1

• **Answer:** a) E9 (Calculation:)

29. An image on the seven segments can be represented by _____ bits.

a) 4 bits b) 7 bits c) 8 bits d) 16 bits

• **Answer:** b) 7 bits, (*Directly from Exam 2025*)

30. Which of the following is NOT a valid representation of a pixel in standard image types? a) 1 bit b) 8 bits c) 12 bits (CMY) d) 24 bits

• **Answer:** c) 12 bits (CMY), (*Based on lecture covering Binary, Gray, and RGB only*)

